

- b) Use Bayes' theorem and part (a) to estimate the probability that an incoming message containing the word w is spam, where $p(w)$ is the probability that w occurs in a spam message and $q(w)$ is the probability that w occurs in a message that is not spam.
23. Suppose that E_1 and E_2 are the events that an incoming mail message contains the words w_1 and w_2 , respectively. Assuming that E_1 and E_2 are independent events and that

$E_1 | S$ and $E_2 | S$ are independent events, where S is the event that an incoming message is spam, and that we have no prior knowledge regarding whether or not the message is spam, show that

$$\begin{aligned} p(S | E_1 \cap E_2) \\ = \frac{p(E_1 | S)p(E_2 | S)}{p(E_1 | S)p(E_2 | S) + p(E_1 | \bar{S})p(E_2 | \bar{S})}. \end{aligned}$$

7.4 Expected Value and Variance

7.4.1 Introduction

The **expected value** of a random variable is the sum over all elements in a sample space of the product of the probability of the element and the value of the random variable at this element. Consequently, the expected value is a weighted average of the values of a random variable. The expected value of a random variable provides a central point for the distribution of values of this random variable. We can solve many problems using the notion of the expected value of a random variable, such as determining who has an advantage in gambling games and computing the average-case complexity of algorithms. Another useful measure of a random variable is its **variance**, which tells us how spread out the values of this random variable are. We can use the variance of a random variable to help us estimate the probability that a random variable takes values far removed from its expected value.

7.4.2 Expected Values



Many questions can be formulated in terms of the value we expect a random variable to take, or more precisely, the average value of a random variable when an experiment is performed a large number of times. Questions of this kind include: How many heads are expected to appear when a coin is flipped 100 times? What is the expected number of comparisons used to find an element in a list using a linear search? To study such questions we introduce the concept of the expected value of a random variable.

Definition 1

The *expected value*, also called the *expectation* or *mean*, of the random variable X on the sample space S is equal to

$$E(X) = \sum_{s \in S} p(s)X(s).$$

The *deviation* of X at $s \in S$ is $X(s) - E(X)$, the difference between the value of X and the mean of X .

Note that when the sample space S has n elements $S = \{x_1, x_2, \dots, x_n\}$, $E(X) = \sum_{i=1}^n p(x_i)X(x_i)$.

Remark: When there are infinitely many elements of the sample space, the expectation is defined only when the infinite series in the definition is absolutely convergent. In particular, the expectation of a random variable on an infinite sample space is finite if it exists.

EXAMPLE 1 Expected Value of a Die Let X be the number that comes up when a fair die is rolled. What is the expected value of X ?

Solution: The random variable X takes the values 1, 2, 3, 4, 5, or 6, each with probability $1/6$. It follows that

$$E(X) = \frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 2 + \frac{1}{6} \cdot 3 + \frac{1}{6} \cdot 4 + \frac{1}{6} \cdot 5 + \frac{1}{6} \cdot 6 = \frac{21}{6} = \frac{7}{2}. \quad \blacktriangleleft$$

EXAMPLE 2 A fair coin is flipped three times. Let S be the sample space of the eight possible outcomes, and let X be the random variable that assigns to an outcome the number of heads in this outcome. What is the expected value of X ?

Extra Examples 

Solution: In Example 10 of Section 7.2 we listed the values of X for the eight possible outcomes when a coin is flipped three times. Because the coin is fair and the flips are independent, the probability of each outcome is $1/8$. Consequently,

$$\begin{aligned} E(X) &= \frac{1}{8} [X(HHH) + X(HHT) + X(HTH) + X(THH) + X(TTH) \\ &\quad + X(THT) + X(HTT) + X(TTT)] \\ &= \frac{1}{8} (3 + 2 + 2 + 2 + 1 + 1 + 1 + 0) = \frac{12}{8} \\ &= \frac{3}{2}. \end{aligned}$$

Consequently, the expected number of heads that come up when a fair coin is flipped three times is $3/2$. 

When an experiment has relatively few outcomes, we can compute the expected value of a random variable directly from its definition, as was done in Example 2. However, when an experiment has a large number of outcomes, it may be inconvenient to compute the expected value of a random variable directly from its definition. Instead, we can find the expected value of a random variable by grouping together all outcomes assigned the same value by the random variable, as Theorem 1 shows.

THEOREM 1

If X is a random variable and $p(X = r)$ is the probability that $X = r$, so that $p(X = r) = \sum_{s \in S, X(s)=r} p(s)$, then

$$E(X) = \sum_{r \in X(S)} p(X = r)r.$$

Proof: Suppose that X is a random variable with range $X(S)$, and let $p(X = r)$ be the probability that the random variable X takes the value r . Consequently, $p(X = r)$ is the sum of the probabilities of the outcomes s such that $X(s) = r$. It follows that

$$E(X) = \sum_{r \in X(S)} p(X = r)r. \quad \blacktriangleleft$$

Example 3 and the proof of Theorem 2 will illustrate the use of this formula. In Example 3 we will find the expected value of the sum of the numbers that appear on two fair dice when they are rolled. In Theorem 2 we will find the expected value of the number of successes when n Bernoulli trials are performed.

EXAMPLE 3 What is the expected value of the sum of the numbers that appear when a pair of fair dice is rolled?

Solution: Let X be the random variable equal to the sum of the numbers that appear when a pair of dice is rolled. In Example 12 of Section 7.2 we listed the value of X for the 36 outcomes of this experiment. The range of X is $\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$. By Example 12 of Section 7.2 we see that

$$\begin{aligned} p(X = 2) &= p(X = 12) = 1/36, \\ p(X = 3) &= p(X = 11) = 2/36 = 1/18, \\ p(X = 4) &= p(X = 10) = 3/36 = 1/12, \\ p(X = 5) &= p(X = 9) = 4/36 = 1/9, \\ p(X = 6) &= p(X = 8) = 5/36, \\ p(X = 7) &= 6/36 = 1/6. \end{aligned}$$

Substituting these values in the formula, we have

$$\begin{aligned} E(X) &= 2 \cdot \frac{1}{36} + 3 \cdot \frac{1}{18} + 4 \cdot \frac{1}{12} + 5 \cdot \frac{1}{9} + 6 \cdot \frac{5}{36} + 7 \cdot \frac{1}{6} \\ &\quad + 8 \cdot \frac{5}{36} + 9 \cdot \frac{1}{9} + 10 \cdot \frac{1}{12} + 11 \cdot \frac{1}{18} + 12 \cdot \frac{1}{36} \\ &= 7. \end{aligned}$$

THEOREM 2

The expected number of successes when n mutually independent Bernoulli trials are performed, where p is the probability of success on each trial, is np .

Proof: Let X be the random variable equal to the number of successes in n trials. By Theorem 2 of Section 7.2 we see that $p(X = k) = C(n, k)p^k q^{n-k}$. Hence, we have

$$\begin{aligned} E(X) &= \sum_{k=1}^n kp(X = k) && \text{by Theorem 1} \\ &= \sum_{k=1}^n kC(n, k)p^k q^{n-k} && \text{by Theorem 2 in Section 7.2} \\ &= \sum_{k=1}^n nC(n-1, k-1)p^k q^{n-k} && \text{by Exercise 21 in Section 6.4} \\ &= np \sum_{k=1}^n C(n-1, k-1)p^{k-1} q^{n-k} && \text{factoring } np \text{ from each term} \\ &= np \sum_{j=0}^{n-1} C(n-1, j)p^j q^{n-1-j} && \text{shifting index of summation with } j = k-1 \\ &= np(p+q)^{n-1} && \text{by the binomial theorem} \\ &= np && \text{because } p+q = 1. \end{aligned}$$

This completes the proof because it shows that the expected number of successes in n mutually independent Bernoulli trials is np . 

We will also show that the hypothesis that the Bernoulli trials are mutually independent in Theorem 2 is not necessary.

7.4.3 Linearity of Expectations

Theorem 3 tells us that expected values are linear. For example, the expected value of the sum of random variables is the sum of their expected values. We will find this property exceedingly useful.

THEOREM 3

If X_i , $i = 1, 2, \dots, n$ with n a positive integer, are random variables on S , and if a and b are real numbers, then

- (i) $E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$
- (ii) $E(aX + b) = aE(X) + b$.

Proof: Part (i) follows for $n = 2$ directly from the definition of expected value, because

$$\begin{aligned} E(X_1 + X_2) &= \sum_{s \in S} p(s)(X_1(s) + X_2(s)) \\ &= \sum_{s \in S} p(s)X_1(s) + \sum_{s \in S} p(s)X_2(s) \\ &= E(X_1) + E(X_2). \end{aligned}$$

The case for n random variables follows easily by mathematical induction using the case of two random variables. (We leave it to the reader to complete the proof.)

To prove part (ii), note that

$$\begin{aligned} E(aX + b) &= \sum_{s \in S} p(s)(aX(s) + b) \\ &= a \sum_{s \in S} p(s)X(s) + b \sum_{s \in S} p(s) \\ &= aE(X) + b \text{ because } \sum_{s \in S} p(s) = 1. \end{aligned}$$



Examples 4 and 5 illustrate how to use Theorem 3.

EXAMPLE 4 Use Theorem 3 to find the expected value of the sum of the numbers that appear when a pair of fair dice is rolled. (This was done in Example 3 without the benefit of this theorem.)

Solution: Let X_1 and X_2 be the random variables with $X_1((i, j)) = i$ and $X_2((i, j)) = j$, so that X_1 is the number appearing on the first die and X_2 is the number appearing on the second die. It is easy to see that $E(X_1) = E(X_2) = 7/2$ because both equal $(1 + 2 + 3 + 4 + 5 + 6)/6 = 21/6 = 7/2$. The sum of the two numbers that appear when the two dice are rolled is the sum $X_1 + X_2$. By Theorem 3, the expected value of the sum is $E(X_1 + X_2) = E(X_1) + E(X_2) = 7/2 + 7/2 = 7$. 

EXAMPLE 5 In the proof of Theorem 2 we found the expected value of the number of successes when n independent Bernoulli trials are performed, where p is the probability of success on each trial by

direct computation. Show how Theorem 3 can be used to derive this result where the Bernoulli trials are not necessarily independent.

Solution: Let X_i be the random variable with $X_i((t_1, t_2, \dots, t_n)) = 1$ if t_i is a success and $X_i((t_1, t_2, \dots, t_n)) = 0$ if t_i is a failure. The expected value of X_i is $E(X_i) = 1 \cdot p + 0 \cdot (1 - p) = p$ for $i = 1, 2, \dots, n$. Let $X = X_1 + X_2 + \dots + X_n$, so that X counts the number of successes when these n Bernoulli trials are performed. Theorem 3, applied to the sum of n random variables, shows that $E(X) = E(X_1) + E(X_2) + \dots + E(X_n) = np$. 

We can take advantage of the linearity of expectations to find the solutions of many seemingly difficult problems. The key step is to express a random variable whose expectation we wish to find as the sum of random variables whose expectations are easy to find. Examples 6 and 7 illustrate this technique.

EXAMPLE 6 Expected Value in the Hatcheck Problem A new employee checks the hats of n people at a restaurant, forgetting to put claim check numbers on the hats. When customers return for their hats, the checker gives them back hats chosen at random from the remaining hats. What is the expected number of hats that are returned correctly?

Solution: Let X be the random variable that equals the number of people who receive the correct hat from the checker. Let X_i be the random variable with $X_i = 1$ if the i th person receives the correct hat and $X_i = 0$ otherwise. It follows that

$$X = X_1 + X_2 + \dots + X_n.$$

Because it is equally likely that the checker returns any of the hats to this person, it follows that the probability that the i th person receives the correct hat is $1/n$. Consequently, by Theorem 1, for all i we have

$$E(X_i) = 1 \cdot p(X_i = 1) + 0 \cdot p(X_i = 0) = 1 \cdot 1/n + 0 = 1/n.$$

By the linearity of expectations (Theorem 3), it follows that

$$E(X) = E(X_1) + E(X_2) + \dots + E(X_n) = n \cdot 1/n = 1.$$

Consequently, the average number of people who receive the correct hat is exactly 1. Note that this answer is independent of the number of people who have checked their hats! (We will find an explicit formula for the probability that no one receives the correct hat in Example 4 of Section 8.6.) 

EXAMPLE 7 Expected Number of Inversions in a Permutation The ordered pair (i, j) is called an **inversion** in a permutation of the first n positive integers if $i < j$ but j precedes i in the permutation. For instance, there are six inversions in the permutation 3, 5, 1, 4, 2; these inversions are

$$(1, 3), (1, 5), (2, 3), (2, 4), (2, 5), (4, 5).$$

Let $I_{i,j}$ be the random variable on the set of all permutations of the first n positive integers with $I_{i,j} = 1$ if (i, j) is an inversion of the permutation and $I_{i,j} = 0$ otherwise. It follows that if X is the random variable equal to the number of inversions in the permutation, then

$$X = \sum_{1 \leq i < j \leq n} I_{i,j}.$$



Note that it is equally likely for i to precede j in a randomly chosen permutation as it is for j to precede i . (To see this, note that there are an equal number of permutations with each of these properties.) Consequently, for all pairs i and j we have

$$E(I_{i,j}) = 1 \cdot p(I_{i,j} = 1) + 0 \cdot p(I_{i,j} = 0) = 1 \cdot 1/2 + 0 = 1/2.$$

Because there are $\binom{n}{2}$ pairs i and j with $1 \leq i < j \leq n$ and by the linearity of expectations (Theorem 3), we have

$$E(X) = \sum_{1 \leq i < j \leq n} E(I_{i,j}) = \binom{n}{2} \cdot \frac{1}{2} = \frac{n(n-1)}{4}.$$

It follows that there are an average of $n(n-1)/4$ inversions in a permutation of the first n positive integers. 

7.4.4 Average-Case Computational Complexity



Links

Computing the average-case computational complexity of an algorithm can be interpreted as computing the expected value of a random variable. Let the sample space of an experiment be the set of possible inputs $a_j, j = 1, 2, \dots, n$, and let X be the random variable that assigns to a_j the number of operations used by the algorithm when given a_j as input. Based on our knowledge of the input, we assign a probability $p(a_j)$ to each possible input value a_j . Then, the average-case complexity of the algorithm is

$$E(X) = \sum_{j=1}^n p(a_j)X(a_j).$$

This is the expected value of X .

Finding the average-case computational complexity of an algorithm is usually much more difficult than finding its worst-case computational complexity, and often involves the use of sophisticated methods. However, there are some algorithms for which the analysis required to find the average-case computational complexity is not difficult. For instance, in Example 8 we will illustrate how to find the average-case computational complexity of the linear search algorithm under different assumptions concerning the probability that the element for which we search is an element of the list.

EXAMPLE 8 Average-Case Complexity of the Linear Search Algorithm We are given a real number x and a list of n distinct real numbers. The linear search algorithm, described in Section 3.1, locates x by successively comparing it to each element in the list, terminating when x is located or when all the elements have been examined and it has been determined that x is not in the list. What is the average-case computational complexity of the linear search algorithm if the probability that x is in the list is p and it is equally likely that x is any of the n elements in the list? (There are $n+1$ possible types of input: one type for each of the n numbers in the list and a last type for numbers not in the list, which we treat as a single input.)

Solution: In Example 4 of Section 3.3 we showed that $2i+1$ comparisons are used if x equals the i th element of the list and, in Example 2 of Section 3.3, we showed that $2n+2$ comparisons are used if x is not in the list. The probability that x equals a_i , the i th element in the list, is

p/n , and the probability that x is not in the list is $q = 1 - p$. It follows that the average-case computational complexity of the linear search algorithm is

$$\begin{aligned}
 E &= \frac{3p}{n} + \frac{5p}{n} + \cdots + \frac{(2n+1)p}{n} + (2n+2)q \\
 &= \frac{p}{n}(3 + 5 + \cdots + (2n+1)) + (2n+2)q \\
 &= \frac{p}{n}((n+1)^2 - 1) + (2n+2)q \\
 &= p(n+2) + (2n+2)q.
 \end{aligned}$$

(The third equality follows from Example 2 of Section 5.1.) For instance, when x is guaranteed to be in the list, we have $p = 1$ (so the probability that $x = a_i$ is $1/n$ for each i) and $q = 0$. Then $E = n + 2$, as we showed in Example 4 in Section 3.3.

When p , the probability that x is in the list, is $1/2$, it follows that $q = 1 - p = 1/2$, so $E = (n+2)/2 + n + 1 = (3n+4)/2$. Similarly, if the probability that x is in the list is $3/4$, we have $p = 3/4$ and $q = 1/4$, so $E = 3(n+2)/4 + (n+1)/2 = (5n+8)/4$.

Finally, when x is guaranteed not to be in the list, we have $p = 0$ and $q = 1$. It follows that $E = 2n + 2$, which is not surprising because we have to search the entire list. 

Example 9 illustrates how the linearity of expectations can help us find the average-case complexity of a sorting algorithm, the insertion sort.

EXAMPLE 9 Average-Case Complexity of the Insertion Sort What is the average number of comparisons used by the insertion sort to sort n distinct elements?

Solution: We first suppose that X is the random variable equal to the number of comparisons used by the insertion sort (described in Section 3.1) to sort a list $a_1, a_2, \dots, a_n$ of n distinct elements. Then $E(X)$ is the average number of comparisons used. (Recall that at step i for $i = 2, \dots, n$, the insertion sort inserts the i th element in the original list into the correct position in the sorted list of the first $i - 1$ elements of the original list.)

We let X_i be the random variable equal to the number of comparisons used to insert a_i into the proper position after the first $i - 1$ elements $a_1, a_2, \dots, a_{i-1}$ have been sorted. Because

$$X = X_2 + X_3 + \cdots + X_n,$$

we can use the linearity of expectations to conclude that

$$E(X) = E(X_2 + X_3 + \cdots + X_n) = E(X_2) + E(X_3) + \cdots + E(X_n).$$

To find $E(X_i)$ for $i = 2, 3, \dots, n$, let $p_j(k)$ denote the probability that the largest of the first j elements in the list occurs at the k th position, that is, that $\max(a_1, a_2, \dots, a_j) = a_k$, where $1 \leq k \leq j$. Because the elements of the list are randomly distributed, it is equally likely for the largest element among the first j elements to occur at any position. Consequently, $p_j(k) = 1/j$. If $X_i(k)$ equals the number of comparisons used by the insertion sort if a_i is inserted into the k th position in the list once $a_1, a_2, \dots, a_{i-1}$ have been sorted, it follows that $X_i(k) = k$. Because it is possible that a_i is inserted in any of the first i positions, we find that

$$E(X_i) = \sum_{k=1}^i p_i(k) \cdot X_i(k) = \sum_{k=1}^i \frac{1}{i} \cdot k = \frac{1}{i} \cdot \sum_{k=1}^i k = \frac{1}{i} \cdot \frac{i(i+1)}{2} = \frac{i+1}{2}.$$

It follows that

$$\begin{aligned} E(X) &= \sum_{i=2}^n E(X_i) = \sum_{i=2}^n \frac{i+1}{2} = \frac{1}{2} \sum_{j=3}^{n+1} j \\ &= \frac{1}{2} \frac{(n+1)(n+2)}{2} - \frac{1}{2}(1+2) = \frac{n^2 + 3n - 4}{4}. \end{aligned}$$

To obtain the third of these equalities we shifted the index of summation, setting $j = i + 1$. To obtain the fourth equality, we used the formula $\sum_{k=1}^m k = m(m+1)/2$ (from Table 2 in Section 2.4) with $m = n + 1$, subtracting off the missing terms with $j = 1$ and $j = 2$. We conclude that the average number of comparisons used by the insertion sort to sort n elements equals $(n^2 + 3n - 4)/4$, which is $\Theta(n^2)$. 

7.4.5 The Geometric Distribution

We now turn our attention to a random variable with infinitely many possible outcomes.

EXAMPLE 10 Suppose that the probability that a coin comes up tails is p . This coin is flipped repeatedly until it comes up tails. What is the expected number of flips until this coin comes up tails?

Links  **Solution:** We first note that the sample space consists of all sequences that begin with any number of heads, denoted by H , followed by a tail, denoted by T . Therefore, the sample space is the set $\{T, HT, HHT, HHHT, HHHHT, \dots\}$. Note that this is an infinite sample space. We can determine the probability of an element of the sample space by noting that the coin flips are independent and that the probability of a head is $1 - p$. Therefore, $p(T) = p$, $p(HT) = (1 - p)p$, $p(HHT) = (1 - p)^2p$, and in general the probability that the coin is flipped n times before a tail comes up, that is, that $n - 1$ heads come up followed by a tail, is $(1 - p)^{n-1}p$. (Exercise 14 asks for a verification that the sum of the probabilities of the points in the sample space is 1.)

Now let X be the random variable equal to the number of flips in an element in the sample space. That is, $X(T) = 1$, $X(HT) = 2$, $X(HHT) = 3$, and so on. Note that $p(X = j) = (1 - p)^{j-1}p$. The expected number of flips until the coin comes up tails equals $E(X)$.

Using Theorem 1, we find that

$$E(X) = \sum_{j=1}^{\infty} j \cdot p(X = j) = \sum_{j=1}^{\infty} j(1 - p)^{j-1}p = p \sum_{j=1}^{\infty} j(1 - p)^{j-1} = p \cdot \frac{1}{p^2} = \frac{1}{p}.$$

[The third equality in this chain follows from Table 2 in Section 2.4, which tells us that $\sum_{j=1}^{\infty} j(1 - p)^{j-1} = 1/(1 - (1 - p))^2 = 1/p^2$.] It follows that the expected number of times the coin is flipped until tails comes up is $1/p$. Note that when the coin is fair we have $p = 1/2$, so the expected number of flips until it comes up tails is $1/(1/2) = 2$. 

The random variable X that equals the number of flips expected before a coin comes up tails is an example of a random variable with a **geometric distribution**.

Definition 2

A random variable X has a *geometric distribution with parameter p* if $p(X = k) = (1 - p)^{k-1}p$ for $k = 1, 2, 3, \dots$, where p is a real number with $0 \leq p \leq 1$.

Geometric distributions arise in many applications because they are used to study the time required before a particular event happens, such as the time required before we find an object with a certain property, the number of attempts before an experiment succeeds, the number of times a product can be used before it fails, and so on.

When we computed the expected value of the number of flips required before a coin comes up tails, we proved Theorem 4.

THEOREM 4

If the random variable X has the geometric distribution with parameter p , then $E(X) = 1/p$.

7.4.6 Independent Random Variables

We have already discussed independent events. We will now define what it means for two random variables to be independent.

Definition 3

The random variables X and Y on a sample space S are *independent* if

$$p(X = r_1 \text{ and } Y = r_2) = p(X = r_1) \cdot p(Y = r_2),$$

or in words, if the probability that $X = r_1$ and $Y = r_2$ equals the product of the probabilities that $X = r_1$ and $Y = r_2$, for all real numbers r_1 and r_2 .

EXAMPLE 11

Are the random variables X_1 and X_2 from Example 4 independent?

Extra Examples ➤

Solution: Let $S = \{1, 2, 3, 4, 5, 6\}$, and let $i \in S$ and $j \in S$. Because there are 36 possible outcomes when the pair of dice is rolled and each is equally likely, we have

$$p(X_1 = i \text{ and } X_2 = j) = 1/36.$$

Furthermore, $p(X_1 = i) = 1/6$ and $p(X_2 = j) = 1/6$, because the probability that i appears on the first die and the probability that j appears on the second die are both $1/6$. It follows that

$$p(X_1 = i \text{ and } X_2 = j) = \frac{1}{36} \quad \text{and} \quad p(X_1 = i)p(X_2 = j) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36},$$

so X_1 and X_2 are independent. ◀

EXAMPLE 12

Show that the random variables X_1 and $X = X_1 + X_2$, where X_1 and X_2 are as defined in Example 4, are not independent.

Solution: Note that $p(X_1 = 1 \text{ and } X = 12) = 0$, because $X_1 = 1$ means the number appearing on the first die is 1, which implies that the sum of the numbers appearing on the two dice cannot equal 12. On the other hand, $p(X_1 = 1) = 1/6$ and $p(X = 12) = 1/36$. Hence $p(X_1 = 1 \text{ and } X = 12) \neq p(X_1 = 1) \cdot p(X = 12)$. This counterexample shows that X_1 and X are not independent. ◀

The expected value of the product of two independent random variables is the product of their expected values, as Theorem 5 shows.

THEOREM 5

If X and Y are independent random variables on a sample space S , then $E(XY) = E(X)E(Y)$.

Proof: To prove this formula, we use the key observation that the event $XY = r$ is the disjoint union of the events $X = r_1$ and $Y = r_2$ over all $r_1 \in X(S)$ and $r_2 \in Y(S)$ with $r = r_1 r_2$. We have

$$\begin{aligned}
 E(XY) &= \sum_{r \in XY(S)} r \cdot p(XY = r) && \text{by Theorem 1} \\
 &= \sum_{r_1 \in X(S), r_2 \in Y(S)} r_1 r_2 \cdot p(X = r_1 \text{ and } Y = r_2) && \text{expressing } XY = r \text{ as a disjoint union} \\
 &= \sum_{r_1 \in X(S)} \sum_{r_2 \in Y(S)} r_1 r_2 \cdot p(X = r_1 \text{ and } Y = r_2) && \text{using a double sum to order the terms} \\
 &= \sum_{r_1 \in X(S)} \sum_{r_2 \in Y(S)} r_1 r_2 \cdot p(X = r_1) \cdot p(Y = r_2) && \text{by the independence of } X \text{ and } Y \\
 &= \sum_{r_1 \in X(S)} (r_1 \cdot p(X = r_1)) \cdot \sum_{r_2 \in Y(S)} r_2 \cdot p(Y = r_2) && \text{by factoring out } r_1 \cdot p(X = r_1) \\
 &= \sum_{r_1 \in X(S)} r_1 \cdot p(X = r_1) \cdot E(Y) && \text{by the definition of } E(Y) \\
 &= E(Y) \left(\sum_{r_1 \in X(S)} r_1 \cdot p(X = r_1) \right) && \text{by factoring out } E(Y) \\
 &= E(Y)E(X) && \text{by the definition of } E(X).
 \end{aligned}$$

We complete the proof by noting that $E(Y)E(X) = E(X)E(Y)$, which is a consequence of the commutative law for multiplication. $\triangleleft$

Note that when X and Y are random variables that are not independent, we cannot conclude that $E(XY) = E(X)E(Y)$, as Example 13 shows.

EXAMPLE 13

Let X and Y be random variables that count the number of heads and the number of tails when a coin is flipped twice. Because $p(X = 2) = 1/4$, $p(X = 1) = 1/2$, and $p(X = 0) = 1/4$, by Theorem 1 we have

$$E(X) = 2 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 0 \cdot \frac{1}{4} = 1.$$

A similar computation shows that $E(Y) = 1$. We note that $XY = 0$ when either two heads and no tails or two tails and no heads come up and that $XY = 1$ when one head and one tail come up. Hence,

$$E(XY) = 1 \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = \frac{1}{2}.$$

It follows that

$$E(XY) \neq E(X)E(Y).$$

This does not contradict Theorem 5 because X and Y are not independent, as the reader should verify (see Exercise 16). ◀

7.4.7 Variance



The expected value of a random variable tells us its average value, but nothing about how widely its values are distributed. For example, if X and Y are the random variables on the set $S = \{1, 2, 3, 4, 5, 6\}$, with $X(s) = 0$ for all $s \in S$ and $Y(s) = -1$ if $s \in \{1, 2, 3\}$ and $Y(s) = 1$ if $s \in \{4, 5, 6\}$, then the expected values of X and Y are both zero. However, the random variable X never varies from 0, while the random variable Y always differs from 0 by 1. The variance of a random variable helps us characterize how widely a random variable is distributed. In particular, it provides a measure of how widely X is distributed about its expected value.

Definition 4

Let X be a random variable on a sample space S . The *variance* of X , denoted by $V(X)$, is

$$V(X) = \sum_{s \in S} (X(s) - E(X))^2 p(s).$$

That is, $V(X)$ is the weighted average of the square of the deviation of X . The *standard deviation* of X , denoted $\sigma(X)$, is defined to be $\sqrt{V(X)}$.

Theorem 6 provides a useful simple expression for the variance of a random variable.

THEOREM 6

If X is a random variable on a sample space S , then $V(X) = E(X^2) - E(X)^2$.

Proof: Note that

$$\begin{aligned} V(X) &= \sum_{s \in S} (X(s) - E(X))^2 p(s) \\ &= \sum_{s \in S} X(s)^2 p(s) - 2E(X) \sum_{s \in S} X(s) p(s) + E(X)^2 \sum_{s \in S} p(s) \\ &= E(X^2) - 2E(X)E(X) + E(X)^2 \\ &= E(X^2) - E(X)^2. \end{aligned}$$

We have used the fact that $\sum_{s \in S} p(s) = 1$ in the next-to-last step. ◀

We can use Theorems 3 and 6 to derive an alternative formula for $V(X)$ that provides some insight into the meaning of the variance of a random variable.

COROLLARY 1

If X is a random variable on a sample space S and $E(X) = \mu$, then $V(X) = E((X - \mu)^2)$.

μ is the Greek letter mu.

Proof: If X is a random variable with $E(X) = \mu$, then

$$\begin{aligned}
 E((X - \mu)^2) &= E(X^2 - 2\mu X + \mu^2) && \text{expanding } (X - \mu)^2 \\
 &= E(X^2) - E(2\mu X) + E(\mu^2) && \text{by part (i) of Theorem 3} \\
 &= E(X^2) - 2\mu E(X) + E(\mu^2) && \text{by part (ii) of Theorem 3, noting that } \mu \text{ is a constant} \\
 &= E(X^2) - 2\mu E(X) + \mu^2 && \text{as } E(\mu^2) = \mu^2, \text{ because } \mu^2 \text{ is a constant} \\
 &= E(X^2) - 2\mu^2 + \mu^2 && \text{because } E(X) = \mu \\
 &= E(X^2) - \mu^2 && \text{simplifying} \\
 &= V(X) && \text{by Theorem 6 and noting that } E(X) = \mu.
 \end{aligned}$$

This completes the proof. 

Corollary 1 tells us that the variance of a random variable X is the expected value of the square of the difference between X and its own expected value. This is commonly expressed as saying that the variance of X is the mean of the square of its deviation. We also say that the standard deviation of X is the square root of the mean of the square of its deviation (often read as the “root mean square” of the deviation).

We now compute the variance of some random variables.

EXAMPLE 14

Extra Examples 

What is the variance of the random variable X with $X(t) = 1$ if a Bernoulli trial is a success and $X(t) = 0$ if it is a failure, where p is the probability of success and q is the probability of failure?

Solution: Because X takes only the values 0 and 1, it follows that $X^2(t) = X(t)$. Hence,

$$V(X) = E(X^2) - E(X)^2 = p - p^2 = p(1 - p) = pq. \quad \text{◀}$$

EXAMPLE 15

Variance of the Value of a Die What is the variance of the random variable X , where X is the number that comes up when a fair die is rolled?

Solution: We have $V(X) = E(X^2) - E(X)^2$. By Example 1 we know that $E(X) = 7/2$. To find $E(X^2)$ note that X^2 takes the values i^2 , $i = 1, 2, \dots, 6$, each with probability $1/6$. It follows that

$$E(X^2) = \frac{1}{6}(1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2) = \frac{91}{6}.$$

We conclude that

$$V(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}. \quad \text{◀}$$

EXAMPLE 16

What is the variance of the random variable $X((i, j)) = 2i$, where i is the number appearing on the first die and j is the number appearing on the second die, when two fair dice are rolled?

Solution: We will use Theorem 6 to find the variance of X . To do so, we need to find the expected values of X and X^2 . Note that because $p(X = k)$ is $1/6$ for $k = 2, 4, 6, 8, 10, 12$ and is 0 otherwise,

$$E(X) = (2 + 4 + 6 + 8 + 10 + 12)/6 = 7,$$

and

$$E(X^2) = (2^2 + 4^2 + 6^2 + 8^2 + 10^2 + 12^2)/6 = 182/3.$$

It follows from Theorem 6 that

$$V(X) = E(X^2) - E(X)^2 = 182/3 - 49 = 35/3.$$

Another useful property is that the variance of the sum of two or more independent random variables is the sum of their variances. The formula that expresses this property is known as **Bienaymé's formula**, after Irenée-Jules Bienaymé, the French mathematician who discovered it in 1853. Bienaymé's formula is useful for computing the variance of the result of n independent Bernoulli trials, for instance.

THEOREM 7

BIENAYMÉ'S FORMULA If X and Y are two independent random variables on a sample space S , then $V(X + Y) = V(X) + V(Y)$. Furthermore, if $X_i, i = 1, 2, \dots, n$, with n a positive integer, are pairwise independent random variables on S , then $V(X_1 + X_2 + \dots + X_n) = V(X_1) + V(X_2) + \dots + V(X_n)$.

Proof: From Theorem 6, we have

$$V(X + Y) = E((X + Y)^2) - E(X + Y)^2.$$

It follows that

$$\begin{aligned} V(X + Y) &= E(X^2 + 2XY + Y^2) - (E(X) + E(Y))^2 \\ &= E(X^2) + 2E(XY) + E(Y^2) - E(X)^2 - 2E(X)E(Y) - E(Y)^2. \end{aligned}$$

Links



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IRENÉE-JULES BIENAYMÉ (1796–1878) Bienaymé, born in Paris, moved with his family to Bruges in 1803 when his father became a government administrator. Bienaymé attended the Lycée impérial in Bruges, and when his family returned to Paris in 1811, the Lycée Louis-le-Grand. As a teenager, he helped defend Paris during the 1814 Napoleonic Wars; in 1815, he became a student at the École Polytechnique. In 1816 he joined the Ministry of Finances to help support his family. In 1819, he left the civil service, taking a job lecturing mathematics at the Académie militaire de Saint-Cyr. Unhappy with conditions there, he soon returned to the Ministry of Finances. He attained the position of inspector general, remaining until 1848, when he was forced to retire when a new regime installed their own people. He was able to return as inspector general in 1850, but he retired a second time in 1852. In 1851 he briefly was professor at the Sorbonne and also served as an expert statistician for Napoleon III.

Bienaymé was one of the founders of the Société Mathématique de France, and in 1875 was its president.

Bienaymé was noted for his ingenuity, but his papers frustrated readers by omitting important proofs. He published sparsely, often in obscure journals. However, he made important contributions to probability and statistics, and to their applications to the social sciences and to finance. Among his important contributions are the Bienaymé-Chebyshev inequality, which provides a simple proof of the law of large numbers, a generalization of Laplace's least square method, and Bienaymé's formula for the variance of a sum of random variables. He studied the extinction of aristocratic families, declining despite general population growth. Bienaymé was a skilled linguist; he translated the works of Chebyshev, a close friend, from Russian to French. It has been suggested that his relative obscurity results from his modesty, his lack of interest in asserting the priority of his discoveries, and the fact that his work was often ahead of its time. He and his brother married two sisters who were daughters of a family friend. Bienaymé and his wife had two sons and three daughters.

Because X and Y are independent, by Theorem 5 we have $E(XY) = E(X)E(Y)$. It follows that

$$\begin{aligned} V(X + Y) &= (E(X^2) - E(X)^2) + (E(Y^2) - E(Y)^2) \\ &= V(X) + V(Y). \end{aligned}$$

We leave the proof of the case for n pairwise independent random variables to the reader (Exercise 34). Such a proof can be constructed by generalizing the proof we have given for the case for two random variables. Note that it is not possible to use mathematical induction in a straightforward way to prove the general case (see Exercise 33). ◀

EXAMPLE 17 Find the variance and standard deviation of the random variable X whose value when two fair dice are rolled is $X((i, j)) = i + j$, where i is the number appearing on the first die and j is the number appearing on the second die.

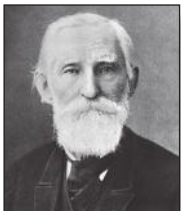
Solution: Let X_1 and X_2 be the random variables defined by $X_1((i, j)) = i$ and $X_2((i, j)) = j$ for a roll of the dice. Then $X = X_1 + X_2$, and X_1 and X_2 are independent, as Example 11 showed. From Theorem 7 it follows that $V(X) = V(X_1) + V(X_2)$. A simple computation as in Example 16, together with Exercise 29 in the Supplementary Exercises, tells us that $V(X_1) = V(X_2) = 35/12$. Hence, $V(X) = 35/12 + 35/12 = 35/6$ and $\sigma(X) = \sqrt{35/6}$. ◀

We will now find the variance of the random variable that counts the number of successes when n independent Bernoulli trials are carried out.

EXAMPLE 18 What is the variance of the number of successes when n independent Bernoulli trials are performed, where, on each trial, p is the probability of success and q is the probability of failure?

Solution: Let X_i be the random variable with $X_i((t_1, t_2, \dots, t_n)) = 1$ if trial t_i is a success and $X_i((t_1, t_2, \dots, t_n)) = 0$ if trial t_i is a failure. Let $X = X_1 + X_2 + \dots + X_n$. Then X counts the number of successes in the n trials. From Theorem 7 it follows that $V(X) = V(X_1) + V(X_2) + \dots + V(X_n)$. Using Example 14 we have $V(X_i) = pq$ for $i = 1, 2, \dots, n$. It follows that $V(X) = npq$. ◀

Links



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PAFNUTY LVOVICH CHEBYSHEV (1821–1894) Chebyshev was born into the gentry in Okatovo, Russia. His father was a retired army officer who had fought against Napoleon. In 1832 the family, with its nine children, moved to Moscow, where Pafnuty completed his high school education at home. He entered the Department of Physics and Mathematics at Moscow University. As a student, he developed a new method for approximating the roots of equations. He graduated from Moscow University in 1841 with a degree in mathematics, and he continued his studies, passing his master's exam in 1843 and completing his master's thesis in 1846.

Chebyshev was appointed in 1847 to a position as an assistant at the University of St. Petersburg. He wrote and defended a thesis in 1847. He became a professor at St. Petersburg in 1860, a position he held until 1882. His book on the theory of congruences written in 1849 was influential in the development of number theory. His work on the distribution of prime numbers was seminal. He proved Bertrand's conjecture that for every integer $n > 3$, there is a prime between n and $2n - 2$. Chebyshev helped develop ideas that

were later used to prove the prime number theorem. Chebyshev's work on the approximation of functions using polynomials is used extensively when computers are used to find values of functions. Chebyshev was also interested in mechanics. He studied the conversion of rotary motion into rectilinear motion by mechanical coupling. The Chebyshev parallel motion is three linked bars approximating rectilinear motion.

7.4.8 Chebyshev's Inequality

How likely is it that a random variable takes a value far from its expected value? Theorem 8, called Chebyshev's inequality, helps answer this question by providing an upper bound on the probability that the value of a random variable differs from the expected value of the random variable by more than a specified amount.

THEOREM 8

Chebyshev's Inequality Let X be a random variable on a sample space S with probability function p . If r is a positive real number, then

$$p(|X(s) - E(X)| \geq r) \leq V(X)/r^2.$$

Proof: Let A be the event

$$A = \{s \in S \mid |X(s) - E(X)| \geq r\}.$$

What we want to prove is that $p(A) \leq V(X)/r^2$. Note that

$$\begin{aligned} V(X) &= \sum_{s \in S} (X(s) - E(X))^2 p(s) \\ &= \sum_{s \in A} (X(s) - E(X))^2 p(s) + \sum_{s \notin A} (X(s) - E(X))^2 p(s). \end{aligned}$$

The second sum in this expression is nonnegative, because each of its summands is nonnegative. Also, because for each element s in A , $(X(s) - E(X))^2 \geq r^2$, the first sum in this expression is at least $\sum_{s \in A} r^2 p(s)$. Hence, $V(X) \geq \sum_{s \in A} r^2 p(s) = r^2 p(A)$. It follows that $V(X)/r^2 \geq p(A)$, so $p(A) \leq V(X)/r^2$, completing the proof. 

EXAMPLE 19

Deviation from the Mean when Counting Tails Suppose that X is the random variable that counts the number of tails when a fair coin is tossed n times. Note that X is the number of successes when n independent Bernoulli trials, each with probability of success $1/2$, are performed. It follows that $E(X) = n/2$ (by Theorem 2) and $V(X) = n/4$ (by Example 18). Applying Chebyshev's inequality with $r = \sqrt{n}$ shows that

$$p(|X(s) - n/2| \geq \sqrt{n}) \leq (n/4)/(\sqrt{n})^2 = 1/4.$$

Consequently, the probability is no more than $1/4$ that the number of tails that come up when a fair coin is tossed n times deviates from the mean by more than $\sqrt{n}$. 

Chebyshev's inequality, although applicable to any random variable, often fails to provide a practical estimate for the probability that the value of a random variable exceeds its mean by a large amount. This is illustrated by Example 20.

EXAMPLE 20

Let X be the random variable whose value is the number appearing when a fair die is rolled. We have $E(X) = 7/2$ (see Example 1) and $V(X) = 35/12$ (see Example 15). Because the only possible values of X are 1, 2, 3, 4, 5, and 6, X cannot take a value more than $5/2$ from its mean, $E(X) = 7/2$. Hence, $p(|X - 7/2| \geq r) = 0$ if $r > 5/2$. By Chebyshev's inequality we know that $p(|X - 7/2| \geq r) \leq (35/12)/r^2$.

For example, when $r = 3$, Chebyshev's inequality tells us that $p(|X - 7/2| \geq 3) \leq (35/12)/9 = 35/108 \approx 0.324$, which is a poor estimate, because $p(|X - 7/2| \geq 3) = 0$. 

Exercises

1. What is the expected number of heads that come up when a fair coin is flipped five times?
 2. What is the expected number of heads that come up when a fair coin is flipped 10 times?
 3. What is the expected number of times a 6 appears when a fair die is rolled 10 times?
 4. A coin is biased so that the probability a head comes up when it is flipped is 0.6. What is the expected number of heads that come up when it is flipped 10 times?
 5. What is the expected sum of the numbers that appear on two dice, each biased so that a 3 comes up twice as often as each other number?
 6. What is the expected value when a \$1 lottery ticket is bought in which the purchaser wins exactly \$10 million if the ticket contains the six winning numbers chosen from the set $\{1, 2, 3, \dots, 50\}$ and the purchaser wins nothing otherwise?
 7. The final exam of a discrete mathematics course consists of 50 true/false questions, each worth two points, and 25 multiple-choice questions, each worth four points. The probability that Linda answers a true/false question correctly is 0.9, and the probability that she answers a multiple-choice question correctly is 0.8. What is her expected score on the final?
 8. What is the expected sum of the numbers that appear when three fair dice are rolled?
 9. Suppose that the probability that x is in a list of n distinct integers is $2/3$ and that it is equally likely that x equals any element in the list. Find the average number of comparisons used by the linear search algorithm to find x or to determine that it is not in the list.
 10. Suppose that we flip a fair coin until either it comes up tails twice or we have flipped it six times. What is the expected number of times we flip the coin?
 11. Suppose that we roll a fair die until a 6 comes up or we have rolled it 10 times. What is the expected number of times we roll the die?
 12. Suppose that we roll a fair die until a 6 comes up.
 - a) What is the probability that we roll the die n times?
 - b) What is the expected number of times we roll the die?
 13. Suppose that we roll a pair of fair dice until the sum of the numbers on the dice is seven. What is the expected number of times we roll the dice?
 14. Show that the sum of the probabilities of a random variable with geometric distribution with parameter p , where $0 < p \leq 1$, equals 1.
 15. Show that if the random variable X has the geometric distribution with parameter p , and j is a positive integer, then $p(X \geq j) = (1 - p)^{j-1}$.
 16. Let X and Y be the random variables that count the number of heads and the number of tails that come up when two fair coins are flipped. Show that X and Y are not independent.
 17. Estimate the expected number of integers with 1000 digits that need to be selected at random to find a prime, if the probability a number with 1000 digits is prime is approximately $1/2302$.
 18. Suppose that X and Y are random variables and that X and Y are nonnegative for all points in a sample space S . Let Z be the random variable defined by $Z(s) = \max(X(s), Y(s))$ for all elements $s \in S$. Show that $E(Z) \leq E(X) + E(Y)$.
 19. Let X be the number appearing on the first die when two fair dice are rolled and let Y be the sum of the numbers appearing on the two dice. Show that $E(X)E(Y) \neq E(XY)$.
 - *20. Show that if $X_1, X_2, \dots, X_n$ are mutually independent random variables, then $E(\prod_{i=1}^n X_i) = \prod_{i=1}^n E(X_i)$.
- The **conditional expectation** of the random variable X given the event A from the sample space S is $E(X|A) = \sum_{r \in X(S)} r \cdot P(X = r|A)$.
21. What is the expected value of the sum of the numbers appearing on two fair dice when they are rolled given that the sum of these numbers is at least nine. That is, what is $E(X|A)$ where X is the sum of the numbers appearing on the two dice and A is the event that $X \geq 9$?
- The **law of total expectation** states that if the sample space S is the disjoint union of the events $S_1, S_2, \dots, S_n$ and X is a random variable, then $E(X) = \sum_{j=1}^n E(X|S_j)P(S_j)$.
22. Prove the law of total expectations.
 23. Use the law of total expectation to find the average weight of a breeding elephant seal, given that 12% of the breeding elephant seals are male and the rest are female, and the expected weights of a breeding elephant seal is 4200 pounds for a male and 1100 pounds for a female.
 24. Let A be an event. Then I_A , the **indicator random variable** of A , equals 1 if A occurs and equals 0 otherwise. Show that the expectation of the indicator random variable of A equals the probability of A , that is, $E(I_A) = P(A)$.
 25. A **run** is a maximal sequence of successes in a sequence of Bernoulli trials. For example, in the sequence $S, S, S, F, S, S, F, F, S$, where S represents success and F represents failure, there are three runs consisting of three successes, two successes, and one success, respectively. Let R denote the random variable on the set of sequences of n independent Bernoulli trials that counts the number of runs in this sequence. Find $E(R)$. [Hint: Show that $R = \sum_{j=1}^n I_j$, where $I_j = 1$ if a run begins at the j th Bernoulli trial and $I_j = 0$ otherwise. Find $E(I_1)$ and then find $E(I_j)$, where $1 < j \leq n$.]
 26. Let $X(s)$ be a random variable, where $X(s)$ is a nonnegative integer for all $s \in S$, and let A_k be the event that $X(s) \geq k$. Show that $E(X) = \sum_{k=1}^{\infty} P(A_k)$.
 27. What is the variance of the number of heads that come up when a fair coin is flipped 10 times?
 28. What is the variance of the number of times a 6 appears when a fair die is rolled 10 times?

29. Let X_n be the random variable that equals the number of tails minus the number of heads when n fair coins are flipped.
- What is the expected value of X_n ?
 - What is the variance of X_n ?
30. Show that if X and Y are independent random variables, then $V(XY) = E(X)^2V(Y) + E(Y)^2V(X) + V(X)V(Y)$.
31. Let $A(X) = E(|X - E(X)|)$, the expected value of the absolute value of the deviation of X , where X is a random variable. Prove or disprove that $A(X + Y) = A(X) + A(Y)$ for all random variables X and Y .
32. Provide an example that shows that the variance of the sum of two random variables is not necessarily equal to the sum of their variances when the random variables are not independent.
33. Suppose that X_1 and X_2 are independent Bernoulli trials each with probability $1/2$, and let $X_3 = (X_1 + X_2) \bmod 2$.
- Show that X_1 , X_2 , and X_3 are pairwise independent, but X_3 and $X_1 + X_2$ are not independent.
 - Show that $V(X_1 + X_2 + X_3) = V(X_1) + V(X_2) + V(X_3)$.
 - Explain why a proof by mathematical induction of Theorem 7 does not work by considering the random variables X_1 , X_2 , and X_3 .
- *34. Prove the general case of Theorem 7. That is, show that if $X_1, X_2, \dots, X_n$ are pairwise independent random variables on a sample space S , where n is a positive integer, then $V(X_1 + X_2 + \dots + X_n) = V(X_1) + V(X_2) + \dots + V(X_n)$. [Hint: Generalize the proof given in Theorem 7 for two random variables. Note that a proof using mathematical induction does not work; see Exercise 33.]
35. Use Chebyshev's inequality to find an upper bound on the probability that the number of tails that come up when a fair coin is tossed n times deviates from the mean by more than $5\sqrt{n}$.
36. Use Chebyshev's inequality to find an upper bound on the probability that the number of tails that come up when a biased coin with probability of heads equal to 0.6 is tossed n times deviates from the mean by more than $\sqrt{n}$.
37. Let X be a random variable on a sample space S such that $X(s) \geq 0$ for all $s \in S$. Show that $p(X(s) \geq a) \leq E(X)/a$ for every positive real number a . This inequality is called **Markov's inequality**.
38. Suppose that the number of cans of soda pop filled in a day at a bottling plant is a random variable with an expected value of 10,000 and a variance of 1000.
- Use Markov's inequality (Exercise 37) to obtain an upper bound on the probability that the plant will fill more than 11,000 cans on a particular day.
 - Use Chebyshev's inequality to obtain a lower bound on the probability that the plant will fill between 9000 and 11,000 cans on a particular day.
39. Suppose that the number of aluminum cans recycled in a day at a recycling center is a random variable with an expected value of 50,000 and a variance of 10,000.
- Use Markov's inequality (Exercise 37) to find an upper bound on the probability that the center will recycle more than 55,000 cans on a particular day.
 - Use Chebyshev's inequality to provide a lower bound on the probability that the center will recycle 40,000 to 60,000 cans on a certain day.
- *40. Suppose the probability that x is the i th element in a list of n distinct integers is $i/[n(n+1)]$. Find the average number of comparisons used by the linear search algorithm to find x or to determine that it is not in the list.
- *41. In this exercise we derive an estimate of the average-case complexity of the variant of the bubble sort algorithm that terminates once a pass has been made with no interchanges. Let X be the random variable on the set of permutations of a set of n distinct integers $\{a_1, a_2, \dots, a_n\}$ with $a_1 < a_2 < \dots < a_n$ such that $X(P)$ equals the number of comparisons used by the bubble sort to put these integers into increasing order.
- Show that, under the assumption that the input is equally likely to be any of the $n!$ permutations of these integers, the average number of comparisons used by the bubble sort equals $E(X)$.
 - Use Example 5 in Section 3.3 to show that $E(X) \leq n(n-1)/2$.
 - Show that the sort makes at least one comparison for every inversion of two integers in the input.
 - Let $I(P)$ be the random variable that equals the number of inversions in the permutation P . Show that $E(X) \geq E(I)$.
 - Let $I_{j,k}$ be the random variable with $I_{j,k}(P) = 1$ if a_k precedes a_j in P and $I_{j,k} = 0$ otherwise. Show that $I(P) = \sum_k \sum_{j < k} I_{j,k}(P)$.
 - Show that $E(I) = \sum_k \sum_{j < k} E(I_{j,k})$.
 - Show that $E(I_{j,k}) = 1/2$. [Hint: Show that $E(I_{j,k}) =$ probability that a_k precedes a_j in a permutation P . Then show it is equally likely for a_k to precede a_j as it is for a_j to precede a_k in a permutation.]
 - Use parts (f) and (g) to show that $E(I) = n(n-1)/4$.
 - Conclude from parts (b), (d), and (h) that the average number of comparisons used to sort n integers is $\Theta(n^2)$.
- *42. In this exercise we find the average-case complexity of the quick sort algorithm, described in the preamble to Exercise 50 in Section 5.4, assuming a uniform distribution on the set of permutations.
- Let X be the number of comparisons used by the quick sort algorithm to sort a list of n distinct integers. Show that the average number of comparisons used by the quick sort algorithm is $E(X)$ (where the sample space is the set of all $n!$ permutations of n integers).
 - Let $I_{j,k}$ denote the random variable that equals 1 if the j th smallest element and the k th smallest element of the initial list are ever compared as the quick sort algorithm sorts the list and equals 0 otherwise. Show that $X = \sum_{k=2}^n \sum_{j=1}^{k-1} I_{j,k}$.

- c) Show that $E(X) = \sum_{k=2}^n \sum_{j=1}^{k-1} p(\text{the } j\text{th smallest element and the } k\text{th smallest element are compared})$.
- d) Show that $p(\text{the } j\text{th smallest element and the } k\text{th smallest element are compared})$, where $k > j$, equals $2/(k-j+1)$.
- e) Use parts (c) and (d) to show that $E(X) = 2(n+1)(\sum_{i=2}^n 1/i) - 2(n-1)$.
- f) Conclude from part (e) and the fact that $\sum_{j=1}^n 1/j \approx \ln n + \gamma$, where $\gamma = 0.57721 \dots$ is Euler's constant, that the average number of comparisons used by the quick sort algorithm is $\Theta(n \log n)$.
- *43. What is the variance of the number of **fixed elements**, that is, elements left in the same position, of a randomly selected permutation of n elements? [Hint: Let X denote the number of fixed points of a random permutation. Write $X = X_1 + X_2 + \dots + X_n$, where $X_i = 1$ if the permutation fixes the i th element and $X_i = 0$ otherwise.]
- The **covariance** of two random variables X and Y on a sample space S , denoted by $\text{Cov}(X, Y)$, is defined to be the expected

value of the random variable $(X - E(X))(Y - E(Y))$. That is, $\text{Cov}(X, Y) = E((X - E(X))(Y - E(Y)))$.

44. Show that $\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$, and use this result to conclude that $\text{Cov}(X, Y) = 0$ if X and Y are independent random variables.
45. Show that $V(X + Y) = V(X) + V(Y) + 2 \text{Cov}(X, Y)$.
46. Find $\text{Cov}(X, Y)$ if X and Y are the random variables with $X((i, j)) = 2i$ and $Y((i, j)) = i + j$, where i and j are the numbers that appear on the first and second of two dice when they are rolled.
47. When m balls are distributed into n bins uniformly at random, what is the probability that the first bin remains empty?
48. What is the expected number of balls that fall into the first bin when m balls are distributed into n bins uniformly at random?
49. What is the expected number of bins that remain empty when m balls are distributed into n bins uniformly at random?

Key Terms and Results

TERMS

sample space: the set of possible outcomes of an experiment

event: a subset of the sample space of an experiment

probability of an event (Laplace's definition): the number of successful outcomes of this event divided by the number of possible outcomes

probability distribution: a function p from the set of all outcomes of a sample space S for which $0 \leq p(x_i) \leq 1$ for $i = 1, 2, \dots, n$ and $\sum_{i=1}^n p(x_i) = 1$, where $x_1, \dots, x_n$ are the possible outcomes

probability of an event E : the sum of the probabilities of the outcomes in E

$p(E|F)$ (conditional probability of E given F): the ratio $p(E \cap F)/p(F)$

independent events: events E and F such that $p(E \cap F) = p(E)p(F)$

pairwise independent events: events $E_1, E_2, \dots, E_n$ such that $p(E_i \cap E_j) = p(E_i)p(E_j)$ for all pairs of integers i and j with $1 \leq j < k \leq n$

mutually independent events: events $E_1, E_2, \dots, E_n$ such that $p(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_m}) = p(E_{i_1})p(E_{i_2}) \dots p(E_{i_m})$ whenever $i_j, j = 1, 2, \dots, m$, are integers with $1 \leq i_1 < i_2 < \dots < i_m \leq n$ and $m \geq 2$

random variable: a function that assigns a real number to each possible outcome of an experiment

distribution of a random variable X : the set of pairs $(r, p(X = r))$ for $r \in X(S)$

uniform distribution: the assignment of equal probabilities to the elements of a finite set

expected value of a random variable: the weighted average of a random variable, with values of the random variable

weighted by the probability of outcomes, that is, $E(X) = \sum_{s \in S} p(s)X(s)$

geometric distribution: the distribution of a random variable X such that $p(X = k) = (1 - p)^{k-1}p$ for $k = 1, 2, \dots$ for some real number p with $0 \leq p \leq 1$

independent random variables: random variables X and Y such that $p(X = r_1 \text{ and } Y = r_2) = p(X = r_1)p(Y = r_2)$ for all real numbers r_1 and r_2

variance of a random variable X : the weighted average of the square of the difference between the value of X and its expected value $E(X)$, with weights given by the probability of outcomes, that is, $V(X) = \sum_{s \in S} (X(s) - E(X))^2 p(s)$

standard deviation of a random variable X : the square root of the variance of X , that is, $\sigma(X) = \sqrt{V(X)}$

Bernoulli trial: an experiment with two possible outcomes

probabilistic (or Monte Carlo) algorithm: an algorithm in which random choices are made at one or more steps

probabilistic method: a technique for proving the existence of objects in a set with certain properties that proceeds by assigning probabilities to objects and showing that the probability that an object has these properties is positive

RESULTS

The probability of exactly k successes when n independent Bernoulli trials are carried out equals $C(n, k)p^k q^{n-k}$, where p is the probability of success and $q = 1 - p$ is the probability of failure.

Bayes' theorem: If E and F are events from a sample space S such that $p(E) \neq 0$ and $p(F) \neq 0$, then

$$p(F | E) = \frac{p(E | F)p(F)}{p(E | F)p(F) + p(E | \bar{F})p(\bar{F})}.$$

$$E(X) = \sum_{r \in X(S)} p(X = r)r.$$

linearity of expectations: $E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n)$ if $X_1, X_2, \dots, X_n$ are random variables.

If X and Y are independent random variables, then $E(XY) = E(X)E(Y)$.

Bienaymé's formula: If $X_1, X_2, \dots, X_n$ are independent random variables, then $V(X_1 + X_2 + \cdots + X_n) = V(X_1) + V(X_2) + \cdots + V(X_n)$.

Chebyshev's inequality: $p(|X(s) - E(X)| \geq r) \leq V(X)/r^2$, where X is a random variable with probability function p and r is a positive real number.

Review Questions

1. a) Define the probability of an event when all outcomes are equally likely.
b) What is the probability that you select the six winning numbers in a lottery if the six different winning numbers are selected from the first 50 positive integers?
2. a) What conditions should be met by the probabilities assigned to the outcomes from a finite sample space?
b) What probabilities should be assigned to the outcome of heads and the outcome of tails if heads comes up three times as often as tails?
3. a) Define the conditional probability of an event E given an event F .
b) Suppose E is the event that when a die is rolled it comes up an even number, and F is the event that when a die is rolled it comes up 1, 2, or 3. What is the probability of F given E ?
4. a) When are two events E and F independent?
b) Suppose E is the event that an even number appears when a fair die is rolled, and F is the event that a 5 or 6 comes up. Are E and F independent?
5. a) What is a random variable?
b) What are the possible values assigned by the random variable X that assigns to a roll of two dice the larger number that appears on the two dice?
6. a) Define the expected value of a random variable X .
b) What is the expected value of the random variable X that assigns to a roll of two dice the larger number that appears on the two dice?
7. a) Explain how the average-case computational complexity of an algorithm, with finitely many possible input values, can be interpreted as an expected value.
b) What is the average-case computational complexity of the linear search algorithm, if the probability that the element for which we search is in the list is $1/3$, and it is equally likely that this element is any of the n elements in the list?
8. a) What is meant by a Bernoulli trial?
b) What is the probability of k successes in n independent Bernoulli trials?
c) What is the expected value of the number of successes in n independent Bernoulli trials?
9. a) What does the linearity of expectations of random variables mean?
b) How can the linearity of expectations help us find the expected number of people who receive the correct hat when a hatter returns hats at random?
10. a) How can probability be used to solve a decision problem, if a small probability of error is acceptable?
b) How can we quickly determine whether a positive integer is prime, if we are willing to accept a small probability of making an error?
11. State Bayes' theorem and use it to find $p(F | E)$ if $p(E | F) = 1/3$, $p(E | \bar{F}) = 1/4$, and $p(F) = 2/3$, where E and F are events from a sample space S .
12. a) What does it mean to say that a random variable has a geometric distribution with parameter p ?
b) What is the mean of a geometric distribution with parameter p ?
13. a) What is the variance of a random variable?
b) What is the variance of a Bernoulli trial with probability p of success?
14. a) What is the variance of the sum of n independent random variables?
b) What is the variance of the number of successes when n independent Bernoulli trials, each with probability p of success, are carried out?
15. What does Chebyshev's inequality tell us about the probability that a random variable deviates from its mean by more than a specified amount?

Supplementary Exercises

1. What is the probability that six consecutive integers will be chosen as the winning numbers in a lottery where each number chosen is an integer between 1 and 40 (inclusive)?
2. In 2012, a player in the Mega Millions lottery picks five different integers between 1 and 56, inclusive, and a sixth integer between 1 and 46, which may duplicate one of the earlier five integers. The player wins the jackpot if the first five numbers picked match the first five numbers drawn and the sixth number matches the sixth number drawn.
a) What is the probability that a player wins the jackpot?

- b) What is the probability that a player wins \$250,000, which is the prize for matching the first five numbers, but not the sixth number, drawn?
 - c) What is the probability that a player wins \$150 by matching exactly three of the first five numbers and the sixth number or by matching four of the first five numbers but not the sixth number?
 - d) What is the probability that a player wins a prize, if a prize is given when the player matches at least three of the first five numbers or the last number.
3. In 2012, a player in the Powerball lottery picks five different integers between 1 and 59, inclusive, and a sixth integer between 1 and 39, which may duplicate one of the earlier five integers. The player wins the jackpot if the first five numbers picked match the first five number drawn and the sixth number matches the sixth number drawn.
- a) What is the probability that a player wins the jackpot?
 - b) What is the probability that a player wins \$200,000, which is the prize for matching the first five numbers, but not the sixth number, drawn?
 - c) What is the probability that a player wins \$100 by matching exactly three of the first five and the sixth numbers or four of the first five numbers but not the sixth number?
 - d) What is the probability that a player wins a prize, if a prize is given when the player matches at least three of the first five numbers or the last number.
4. What is the probability that a hand of 13 cards contains no pairs?
5. What is the probability that a 13-card bridge hand contains
- a) all 13 hearts?
 - b) 13 cards of the same suit?
 - c) seven spades and six clubs?
 - d) seven cards of one suit and six cards of a second suit?
 - e) four diamonds, six hearts, two spades, and one club?
 - f) four cards of one suit, six cards of a second suit, two cards of a third suit, and one card of the fourth suit?
6. What is the probability that a seven-card poker hand contains
- a) four cards of one kind and three cards of a second kind?
 - b) three cards of one kind and pairs of each of two different kinds?
 - c) pairs of each of three different kinds and a single card of a fourth kind?
 - d) pairs of each of two different kinds and three cards of a third, fourth, and fifth kind?
 - e) cards of seven different kinds?
 - f) a seven-card flush?
 - g) a seven-card straight?
 - h) a seven-card straight flush?

An **octahedral die** has eight faces that are numbered 1 through 8.

- 7. a) What is the expected value of the number that comes up when a fair octahedral die is rolled?

- b) What is the variance of the number that comes up when a fair octahedral die is rolled?

A **dodecahedral die** has 12 faces that are numbered 1 through 12.

- 8. a) What is the expected value of the number that comes up when a fair dodecahedral die is rolled?
 - b) What is the variance of the number that comes up when a fair dodecahedral die is rolled?
9. Suppose that a pair of fair octahedral dice is rolled.
- a) What is the expected value of the sum of the numbers that come up?
 - b) What is the variance of the sum of the numbers that come up?
10. Suppose that a pair of fair dodecahedral dice is rolled.
- a) What is the expected value of the sum of the numbers that come up?
 - b) What is the variance of the sum of the numbers that come up?
11. Suppose that a fair standard (cubic) die and a fair octahedral die are rolled together.
- a) What is the expected value of the sum of the numbers that come up?
 - b) What is the variance of the sum of the numbers that come up?
12. Suppose that a fair octahedral die and a fair dodecahedral die are rolled together.
- a) What is the expected value of the sum of the numbers that come up?
 - b) What is the variance of the sum of the numbers that come up?
13. Suppose n people, $n \geq 3$, play “odd person out” to decide who will buy the next round of refreshments. The n people each flip a fair coin simultaneously. If all the coins but one come up the same, the person whose coin comes up different buys the refreshments. Otherwise, the people flip the coins again and continue until just one coin comes up different from all the others.
- a) What is the probability that the odd person out is decided in just one coin flip?
 - b) What is the probability that the odd person out is decided with the k th flip?
 - c) What is the expected number of flips needed to decide odd person out with n people?
14. Suppose that p and q are primes and $n = pq$. What is the probability that a randomly chosen positive integer less than n is not divisible by either p or q ?
- *15. Suppose that m and n are positive integers. What is the probability that a randomly chosen positive integer less than mn is not divisible by either m or n ?
16. Suppose that $E_1, E_2, \dots, E_n$ are n events with $p(E_i) > 0$ for $i = 1, 2, \dots, n$. Show that

$$\begin{aligned}
 p(E_1 \cap E_2 \cap \dots \cap E_n) \\
 &= p(E_1)p(E_2 | E_1)p(E_3 | E_1 \cap E_2) \\
 &\quad \dots p(E_n | E_1 \cap E_2 \cap \dots \cap E_{n-1}).
 \end{aligned}$$

17. There are three cards in a box. Both sides of one card are black, both sides of one card are red, and the third card has one black side and one red side. We pick a card at random and observe only one side.
- If the side is black, what is the probability that the other side is also black?
 - What is the probability that the opposite side is the same color as the one we observed?
18. What is the probability that when a fair coin is flipped n times an equal number of heads and tails appear?
19. What is the probability that a randomly selected bit string of length 10 is a palindrome?
20. What is the probability that a randomly selected bit string of length 11 is a palindrome?
21. Consider the following game. A person flips a coin repeatedly until a head comes up. This person receives a payment of 2^n dollars if the first head comes up at the n th flip.
- Let X be a random variable equal to the amount of money the person wins. Show that the expected value of X does not exist (that is, it is infinite). Show that a rational gambler, that is, someone willing to pay to play the game as long as the price to play is not more than the expected payoff, should be willing to wager any amount of money to play this game. (This is known as the **St. Petersburg paradox**. Why do you suppose it is called a paradox?)
 - Suppose that the person receives 2^n dollars if the first head comes up on the n th flip where $n < 8$ and $2^8 = 256$ dollars if the first head comes up on or after the eighth flip. What is the expected value of the amount of money the person wins? How much money should a person be willing to pay to play this game?
22. Suppose that n balls are tossed into b bins so that each ball is equally likely to fall into any of the bins and that the tosses are independent.
- Find the probability that a particular ball lands in a specified bin.
 - What is the expected number of balls that land in a particular bin?
 - What is the expected number of balls tossed until a particular bin contains a ball?
- *d) What is the expected number of balls tossed until all bins contain a ball? [Hint: Let X_i denote the number of tosses required to have a ball land in an i th bin once $i - 1$ bins contain a ball. Find $E(X_i)$ and use the linearity of expectations.]
23. Suppose that A and B are events with probabilities $p(A) = 3/4$ and $p(B) = 1/3$.
- What is the largest $p(A \cap B)$ can be? What is the smallest it can be? Give examples to show that both extremes for $p(A \cap B)$ are possible.
 - What is the largest $p(A \cup B)$ can be? What is the smallest it can be? Give examples to show that both extremes for $p(A \cup B)$ are possible.
24. Suppose that A and B are events with probabilities $p(A) = 2/3$ and $p(B) = 1/2$.
- What is the largest $p(A \cap B)$ can be? What is the smallest it can be? Give examples to show that both extremes for $p(A \cap B)$ are possible.
 - What is the largest $p(A \cup B)$ can be? What is the smallest it can be? Give examples to show that both extremes for $p(A \cup B)$ are possible.
25. Recall from Definition 5 in Section 7.2 that the events $E_1, E_2, \dots, E_n$ are **mutually independent** if $p(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_m}) = p(E_{i_1})p(E_{i_2}) \dots p(E_{i_m})$ whenever $i_j, j = 1, 2, \dots, m$, are integers with $1 \leq i_1 < i_2 < \dots < i_m \leq n$ and $m \geq 2$.
- Write out the conditions required for three events E_1, E_2 , and E_3 to be mutually independent.
 - Let E_1, E_2 , and E_3 be the events that the first flip comes up heads, that the second flip comes up tails, and that the third flip comes up tails, respectively, when a fair coin is flipped three times. Are E_1, E_2 , and E_3 mutually independent?
 - Let E_1, E_2 , and E_3 be the events that the first flip comes up heads, that the third flip comes up heads, and that an even number of heads come up, respectively, when a fair coin is flipped three times. Are E_1, E_2 , and E_3 pairwise independent? Are they mutually independent?
 - Let E_1, E_2 , and E_3 be the events that the first flip comes up heads, that the third flip comes up heads, and that exactly one of the first flip and third flip come up heads, respectively, when a fair coin is flipped three times. Are E_1, E_2 , and E_3 pairwise independent? Are they mutually independent?
 - How many conditions must be checked to show that n events are mutually independent?
26. Suppose that A and B are events from a sample space S such that $p(A) \neq 0$ and $p(B) \neq 0$. Show that if $p(B | A) < p(B)$, then $p(A | B) < p(A)$.
- In Exercise 27 we consider the **two children problem**, introduced in 1959 by Martin Gardner in his Mathematical Games column in *Scientific American*. A version of the puzzle asks: "We meet Mr. Smith as he is walking down the street with a young child whom he introduces as his son. He also tells us that he has two children. What is the probability that his other child is a son?" We will show that this puzzle is ambiguous, leading to a paradox, by showing that there are two reasonable answers to this problem and we will describe how to make the puzzle unambiguous.
- *27. a) Solve this puzzle in two different ways. First, answer the problem by considering the probability of the gender of the second child. Then, determine the probability differently, by considering the four different possibilities for a family of two children.
- b) Show that the answer to the puzzle becomes unambiguous if we also know that Mr. Smith chose his walking companion at random from his two children.

- c) Another variation of this puzzle asks “When we meet Mr. Smith, he tells us that he has two children and at least one is a son. What is the probability that his other child is a son?” Solve this variation of the puzzle, explaining why it is unambiguous.
28. In 2010, the puzzle designer Gary Foshee posed this problem: “Mr. Smith has two children, one of whom is a son born on a Tuesday. What is the probability that Mr. Smith has two sons?” Show that there are two different answers to this puzzle, depending on whether Mr. Smith specifically mentioned his son because he was born on a Tuesday or whether he randomly chose a child and reported its gender and birth day of the week. [Hint: For the first possibility, enumerate all the equally likely possibilities for the gender and birth day of the week of the other child. To do, this consider first the cases where the older child is a boy born on a Tuesday and then the case where the older child is not a boy born on a Tuesday.]
29. Let X be a random variable on a sample space S . Show that $V(aX + b) = a^2 V(X)$ whenever a and b are real numbers.
30. Use Chebyshev’s inequality to show that the probability that more than 10 people get the correct hat back when a hatcheck person returns hats at random does not exceed $1/100$ no matter how many people check their hats. [Hint: Use Example 6 and Exercise 43 in Section 7.4.]
31. Suppose that at least one of the events $E_j, j = 1, 2, \dots, m$, is guaranteed to occur and no more than two can occur. Show that if $p(E_j) = q$ for $j = 1, 2, \dots, m$ and $p(E_j \cap E_k) = r$ for $1 \leq j < k \leq m$, then $q \geq 1/m$ and $r \leq 2/m$.
32. Show that if m is a positive integer, then the probability that the m th success occurs on the $(m + n)$ th trial when independent Bernoulli trials, each with probability p of success, are run, is $\binom{n+m-1}{n} q^n p^m$.
33. There are n different types of collectible cards you can get as prizes when you buy a particular product. Suppose that every time you buy this product it is equally likely that you get any type of these cards. Let X be the random variable equal to the number of products that need to be purchased to obtain at least one of each type of card and let X_j be the random variable equal to the number of additional products that must be purchased after j different cards have been collected until a new card is obtained for $j = 0, 1, \dots, n - 1$.
- Show that $X = \sum_{j=0}^{n-1} X_j$.
 - Show that after j distinct types of cards have been obtained, the card obtained with the next purchase will be a card of a new type with probability $(n - j)/n$.
 - Show that X_j has a geometric distribution with parameter $(n - j)/n$.
 - Use parts (a) and (c) to show that $E(X) = n \sum_{j=1}^n 1/j$.
 - Use the approximation $\sum_{j=1}^n 1/j \approx \ln n + \gamma$, where $\gamma = 0.57721 \dots$ is Euler’s constant, to find the expected number of products that you need to buy to get one card of each type if there are 50 different types of cards.

34. The **maximum satisfiability problem** asks for an assignment of truth values to the variables in a compound proposition in conjunctive normal form (which expresses a compound proposition as the conjunction of clauses where each clause is the disjunction of two or more variables or their negations) that makes as many of these clauses true as possible. For example, three but not four of the clauses in

$$(p \vee q) \wedge (p \vee \neg q) \wedge (\neg p \vee r) \wedge (\neg p \vee \neg r)$$

can be made true by an assignment of truth values to p, q , and r . We will show that probabilistic methods can provide a lower bound for the number of clauses that can be made true by an assignment of truth values to the variables.

- Suppose that there are n variables in a compound proposition in conjunctive normal form. If we pick a truth value for each variable randomly by flipping a coin and assigning true to the variable if the coin comes up heads and false if it comes up tails, what is the probability of each possible assignment of truth values to the n variables?
 - Assuming that each clause is the disjunction of exactly two distinct variables or their negations, what is the probability that a given clause is true, given the random assignment of truth values from part (a)?
 - Suppose that there are D clauses in the compound proposition. What is the expected number of these clauses that are true, given the random assignment of truth values of the variables?
 - Use part (c) to show that for every compound proposition in conjunctive normal form there is an assignment of truth values to the variables that makes at least $3/4$ of the clauses true.
35. What is the probability that each player has a hand containing an ace when the 52 cards of a standard deck are dealt to four players?
- *36. The following method can be used to generate a random permutation of a sequence of n terms. First, interchange the n th term and the $r(n)$ th term where $r(n)$ is a randomly selected integer with $1 \leq r(n) \leq n$. Next, interchange the $(n - 1)$ st term of the resulting sequence with its $r(n - 1)$ st term where $r(n - 1)$ is a randomly selected integer with $1 \leq r(n - 1) \leq n - 1$. Continue this process until $j = n$, where at the j th step you interchange the $(n - j + 1)$ st term of the resulting sequence with its $r(n - j + 1)$ st term, where $r(n - j + 1)$ is a randomly selected integer with $1 \leq r(n - j + 1) \leq n - j + 1$. Show that when this method is followed, each of the $n!$ different permutations of the terms of the sequence is equally likely to be generated. [Hint: Use mathematical induction, assuming that the probability that each of the permutations of $n - 1$ terms produced by this procedure for a sequence of $n - 1$ terms is $1/(n - 1)!.$]

Computer Projects

Write programs with these input and output.

1. Given a real number p with $0 \leq p \leq 1$, generate random numbers taken from a Bernoulli distribution with probability p .
2. Given a positive integer n , generate a random permutation of the set $\{1, 2, 3, \dots, n\}$. (See Exercise 36 in the Supplementary Exercises.)
3. Given positive integers m and n , generate m random permutations of the first n positive integers. Find the number of inversions in each permutation and determine the average number of these inversions.
4. Given a positive integer n , simulate n repeated flips of a biased coin with probability p of heads and determine the number of heads that come up. Display the cumulative results.
5. Given positive integers n and m , generate m random permutations of the first n positive integers. Sort each permutation using the insertion sort, counting the number of comparisons used. Determine the average number of comparisons used over all m permutations.
6. Given positive integers n and m , generate m random permutations of the first n positive integers. Sort each permutation using the version of the bubble sort that terminates when a pass has been made with no interchanges, counting the number of comparisons used. Determine the average number of comparisons used over all m permutations.
7. Given a positive integer m , simulate the collection of cards that come with the purchase of products to find the number of products that must be purchased to obtain a full set of m different collector cards. (See Supplementary Exercise 33.)
8. Given positive integers m and n , simulate the placement of n keys, where a record with key k is placed at location $h(k) = k \bmod m$ and determine whether there is at least one collision.
9. Given a positive integer n , find the probability of selecting the six integers from the set $\{1, 2, \dots, n\}$ that were mechanically selected in a lottery.
10. Simulate repeated trials of the Monty Hall Three-Door problem (Example 10 in Section 7.1) to calculate the probability of winning with each strategy.
11. Given a list of words and the empirical probabilities they occur in spam e-mails and in e-mails that are not spam, determine the probability that a new e-mail message is spam.

Computations and Explorations

Use a computational program or programs you have written to do these exercises.

1. Find the probabilities of each type of hand in five-card poker and rank the types of hands by their probability.
2. Find some conditions such that the expected value of buying a \$1 lottery ticket in the New Jersey Pick-6 lottery has an expected value of more than \$1. To win you have to select the six numbers drawn, where order does not matter, from the positive integers 1 to 49, inclusive. The winnings are split evenly among holders of winning tickets. Be sure to consider the total size of the pot going into the drawing and the number of people buying tickets.
3. Estimate the probability that two integers selected at random are relatively prime by testing a large number of randomly selected pairs of integers. Look up the theorem that gives this probability and compare your results with the correct probability.
4. Determine the number of people needed to ensure that the probability at least two of them have the same day of the year as their birthday is at least 70%, at least 80%, at least 90%, at least 95%, at least 98%, and at least 99%.
5. Generate a list of 100 randomly selected permutations of the set of the first 100 positive integers. (See Exercise 36 in the Supplementary Exercises.)
6. Given a collection of e-mail messages, each determined to be spam or not to be spam, develop a Bayesian filter based on the appearance of particular words in these messages.
7. Simulate the odd-person-out procedure (described in Exercise 13 of the Supplementary Exercises) for n people with $3 \leq n \leq 10$. Run a large number of trials for each value of n and use the results to estimate the expected number of flips needed to find the odd person out. Does your result agree with that found in Exercise 29 in Section 7.2? Vary the problem by supposing that exactly one person has a biased coin with probability of heads $p \neq 0.5$.
8. Given a positive integer n , simulate a hatter randomly giving hats back to people. Determine the number of people who get the correct hat back.

Writing Projects

Respond to these with essays using outside sources.

1. Describe the origins of probability theory and the first uses of this theory, including those by Cardano, Pascal, and Laplace.
2. Describe the different bets you can make when you play roulette. Find the probability of each of these bets in the American version where the wheel contains the numbers 0 and 00. Which is the best bet and which is the worst for you?
3. Discuss the probability of winning when you play the game of blackjack versus a casino. Is there a winning strategy for the person playing against the house?
4. Investigate the game of craps and discuss the probability that the shooter wins and how close to a fair game it is.
5. Discuss issues involved in developing successful spam filters and the current situation in the war between spammers and people trying to filter spam out.
6. Discuss the history and solution of what is known as the Newton–Pepys problem, which asks which is most likely: rolling at least one six when six dice are rolled, rolling at least two sixes when 12 dice are rolled, or rolling at least three sixes when 18 dice are rolled.
7. Explain how Erdős and Rényi first used the probabilistic method and describe some other applications of this method.
8. Discuss the different types of probabilistic algorithms and describe some examples of each type.